3.1 Translations

A translation slides every point of a figure the same distance in the same direction, and that single shared shift is why translations are rigid motions — distance, angle measure, and orientation all survive intact. This section goes past applying a rule to one point: it asks you to prove algebraically why distance is preserved, translate an entire triangle vertex by vertex, work backward from an image to a preimage, and combine two translations into one equivalent vector.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. $(3, 1)$ by $\langle 2, 5\rangle$?

2. $(x, y) \to (x - 3, y + 4)$; image of $(4, 0)$?

3. $(0, 0)$ to $(3, -2)$. Vector?

4. Is a translation a rigid motion?

5. Does translation reverse orientation?

6. $(-1, 5)$ by $\langle 4, -3\rangle$?

7. $(x, y) \to (x + 7, y)$ moves how?

8. $(x, y) \to (x, y - 2)$ moves how?

9. Rule $(x + 1, y - 3)$; image $(2, 4)$. Preimage?

10. $A(1, 1)$ by $\langle -2, 3\rangle$?

11. $(4, 7)$ to $(4, 1)$. Vector?

12. Is a figure congruent to its translate?

13. Translate left 5, up 2: rule?

14. Triangle $D(-1, 2)$, $E(3, 2)$, $F(1, -2)$ is translated by $\langle -2, 5\rangle$. Find all three image vertices.

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15. Segment endpoints $(2, -3)$ and $(6, 1)$ have midpoint $(4, -1)$. Translate by $\langle -5, 2 \rangle$. Find the new midpoint (a) by translating the midpoint directly and (b) by translating both endpoints and averaging. Confirm they match.

✓ Answer Key

1. $(5, 6)$
2. $(1, 4)$
3. $\langle 3, -2 \rangle$
4. Yes
5. No
6. $(3, 2)$
7. Right 7
8. Down 2
9. $(1, 7)$
10. $(-1, 4)$
11. $\langle 0, -6 \rangle$
12. Yes
13. $(x, y) \to (x - 5, y + 2)$
14. $D'(-3, 7), E'(1, 7), F'(-1, 3)$
15. Both methods give $(-1, 1)$